

Majeti Lahari

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Professional Summary

Software Developer with 3+ years of experience at IBM, building and deploying production-grade end-to-end web applications, GenAI, RAG, and AI-driven systems on cloud infrastructure. Proven ability to automate large-scale workflows, debug complex production issues, perform incident resolution and root cause analysis, develop proof-of-concept (POC) solutions, improve system reliability, and deliver measurable business impact. Experienced in leading teams, mentoring engineers, and collaborating across product, design and infrastructure functions.

Skills

Backend & Frameworks: Python, Flask, LangChain, FastAPI, PHP (Laravel), RESTful APIs

Cloud & DevOps: IBM Cloud, OpenShift, Docker, Version Control (Git), CI/CD Pipelines (Jenkins, Tekton, Cirrus), Secrets Manager, Cloud Object Storage, Shell Scripting

Databases & Vector Stores: IBM Db2, Milvus, ChromaDB, SQL

Testing: Unit Testing, Integration Testing, Test Automation

Core CS: Object-Oriented Programming, Data Structures & Algorithms

Frontend: HTML5, CSS3, Angular, React, TypeScript, Streamlit, Responsive UI Design

AI / ML Integration: LLMs (Watsonx.ai, OpenAI, Hugging Face, Gemini), RAG, Agentic AI, Natural Language Processing, Machine Learning, Deep Learning

Security & Quality: Mend, Twistlock, ITSS-compliant Pipelines, Security Scanning, Performance Optimization

Professional Skills: Agile Delivery, Technical Leadership, Mentorship, Application Monitoring and Production Support, Root Cause Analysis, Incident Resolution, Debugging, Requirement Analysis, POC Development, Cross-functional Collaboration, Executive Communication

Work Experience

IBM India Private Limited, Bangalore

Full Stack Software Developer (DevOps)

Apr 2025 - Present

- Designed and deployed end-to-end RAG-based production chatbot on IBM Cloud using Flask, Docker, Jenkins CI/CD, and Milvus vector search integrated with Watsonx.ai LLMs, enabling instant access to 100% of project documentation and reducing manual data retrieval by 90%.
- Implemented secure, ITSS-compliant CI/CD pipelines with Mend and Twistlock security scanning, publishing audit-ready reports to Cloud Object Storage.

- Automated 10+ operational workflows, monitoring tasks and incident-handling activities using shell scripting, saving 30+ hours per week and reducing operational errors by 30%.
- Designed and implemented an AI-driven automated handover communication system integrating Watsonx Orchestrate, Python APIs, ServiceNow incident data, Box-hosted operational reports, and Outlook. Replaced manual daily handover creation across global operations with a fully automated workflow, reducing preparation time from 20–30 minutes per handover to under a minute, saving 250+ hours annually while improving reporting accuracy and operational visibility.
- Architected full stack automated dashboards using Angular and PHP, providing real-time feeder and process visibility, eliminating manual tracking and enabling one-click operational insights for application and leadership teams.
- Created multiple rapid proof-of-concepts (POCs) and prototype solutions for emerging business requirements, automation initiatives, and AI-driven use cases, enabling faster requirement validation and reducing implementation uncertainty for development teams.
- Led vulnerability remediation initiatives by analyzing Mend and Twistlock security reports, investigating CVEs through GitHub Advisory Database and security advisories, coordinating dependency and container-image fixes, and serving as the primary point of contact for vulnerability-related support, reducing security compliance risks and accelerating remediation turnaround times.
- Demonstrated technical leadership by conducting knowledge transfers, leading debugging discussions, reviewing code, mentoring 5 interns, and driving delivery priorities to improve onboarding efficiency and overall team productivity.

Full Stack Software Developer

Jan 2023 – Apr 2025

- Developed, unit tested, deployed, and maintained Python-based user-centric web applications and RESTful APIs with clear, context-aware exception handling, reducing user errors by 30% and improving overall UX.
- Led an 11-member team to build a Judges' Award-winning multi-agent LLM system at IBM Developers Jumpstart, reducing manual code review effort by 60% and accelerating approval cycles.
- Engineered Excel-driven batch automation for Sales role creation, reducing processing time from minutes per role to minutes for hundreds of roles (99% time reduction). Project won 2nd Place in 1Q25 Client Advocacy Awards.
- Created a natural-language chatbot integrating IBM Db2 with LLM summarization to deliver real-time system health insights, incident triaging, and operational insights, improving incident response times by 40%.
- Built an AI-assisted incident resolution platform leveraging historical data of recurring production and operational issues to suggest probable root causes and fixes, improving resolution efficiency by 25%.
- Implemented a prompt-driven GenAI system to auto-generate unit and integration test cases, increasing test coverage by 30% and reducing manual QA effort.
- Performed deep production debugging, log analysis, and root cause investigations for critical application and infrastructure issues, enabling faster incident resolution and improved platform stability.
- Maintained technical documentation, deployment workflows, and operational procedures and designed SQL queries and optimized database interactions for application workflows and reporting.

Project Work

LLM-Powered Multimodal Quiz Generator

- Built an end-to-end multimodal quiz generation platform supporting text, documents, images, audio, video, and URLs via a unified processing pipeline.

- Implemented adaptive MCQ generation and validation using LLMs, OCR, audio/video transcription, and modular pipeline architecture.
- Tech Stack: Mistral LLM, Prompt Engineering, OCR, Whisper (Audio-to-Text), MoviePy (Video-to-Text), PyMuPDF, python-docx, BeautifulSoup, Python, Modular Pipeline Architecture, Streamlit.

AI Portfolio Assistant (RAG-Based)

- Integrated a RAG-powered AI assistant into a personal portfolio, enabling natural-language exploration of content with strict grounding rules to prevent hallucinations.
- Deployed locally and on cloud using Flask, vector databases, and LLM inference pipelines for reliable recruiter-facing interactions.
- Tech Stack: Llama 3.1 (8B), Sentence Transformers (all-MiniLM-L6-v2), RAG Architecture, Python, Flask, FlaskCORS, ChromaDB, Ollama (Local deployment), Playwright, React, JavaScript, CSS3, Render, Vercel.

Education

Ramachandra College of Engineering, Eluru | JNTU-Kakinada

Aug 2019 – May 2023

B.Tech in Computer Science and Engineering

CGPA: 8.14/10

Relevant Coursework: Object Oriented Programming, Databases, Data Structures and Algorithms, Operating Systems, Computer Networks, Machine Learning, Deep Learning, Artificial Intelligence, Data Mining, NLP, Advanced Data Structures and Algorithms, Information Retrieval, Image Processing.

Awards and Recognition

- **1st Place & Judges' Award** - IBM Developers Jumpstart (Multi-Agent LLM System).
- **2nd Place** - 1Q25 Client Advocacy Awards (Sales Plan Automation).

Early Projects and Certifications

- Named Entity Recognition using Bi-LSTM.
- Machine Learning-Based Housing Price Prediction.
- Certified Machine Learning Intern at Pantechelearning.